

Calculus I (Track A)

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Chapter 1

Basics of logic and set theory

1.1 Propositions and connectives

We call a *proposition* a declarative statement to which we can assign a truth value (i.e. 1 for “true” and 0 for “false”).

Example 1.1.1. *The mathematical expression $x < 0$ is not a proposition. The mathematical expression $1 < 2$ is a proposition.*

We will consider propositions about which we can always determine whether they are true or false. This is an idealisation: a conjecture is a statement believed to be a proposition in this sense, but whose truth value is not currently known, and by Gödel’s incompleteness theorems any sufficiently rich axiom system contains statements that can be neither proved nor disproved within it.

From the point of view of logic, only whether a given proposition is true or false is essential. The fact that a proposition p is true is written as $p = 1$, and when it is false we write $p = 0$. If $p = 1$, we say that p has logical value 1; if $p = 0$, then p has logical value 0.

The negation of a proposition p is denoted by $\neg p$. Obviously, $p = 0$ if and only if $\neg p = 1$. We can create more complex propositions using connectives such as $\neg, \vee, \wedge, \Rightarrow, \Leftrightarrow$. The basic connectives are:

1. **Disjunction** $p \vee q$ (p or q).
2. **Conjunction** $p \wedge q$ (p and q).
3. **Implication** $p \Rightarrow q$ (p implies q ; if p , then q).
4. **Equivalence** $p \Leftrightarrow q$ (p is equivalent to q ; p if and only if q).

They can be defined using so-called truth tables:

p	q	$\neg p$	$p \vee q$	$p \wedge q$	$p \Rightarrow q$	$p \Leftrightarrow q$
0	0	1	0	0	1	1
0	1	1	1	0	1	0
1	0	0	1	0	0	0
1	1	0	1	1	1	1

Remark 1.1.2 (Two levels of equivalence). The symbol $\Leftrightarrow$ above is a connective: it builds a new proposition $p \Leftrightarrow q$ out of p and q . We reserve the symbol $\iff$ for statements about propositions, namely that two expressions have the same truth value for every assignment of truth values to their variables. Thus “ $p \Leftrightarrow q$ ” is a proposition, while “ $p \iff q$ ” asserts something about p and q ; the two agree in the sense that $p \iff q$ holds exactly when $p \Leftrightarrow q$ has logical value 1 in every row of the truth table.

Example 1.1.3. The proposition $2 = -2$ is false. Nevertheless, it implies the following true proposition:

$$4 = 4.$$

Theorem 1.1.4 (De Morgan’s laws). Let p and q be propositions. Then

$$\neg(p \vee q) \iff (\neg p) \wedge (\neg q),$$

$$\neg(p \wedge q) \iff (\neg p) \vee (\neg q).$$

In other words, $\neg(p \vee q)$ and $(\neg p) \wedge (\neg q)$ have the same truth value, and so do $\neg(p \wedge q)$ and $(\neg p) \vee (\neg q)$.

Proof. We will prove each law by constructing truth tables. In the tables below, 1 denotes true and 0 denotes false. Two logical expressions are equivalent if they have identical truth values for every possible combination of p and q .

Law 1: $\neg(p \vee q) \iff (\neg p) \wedge (\neg q)$.

The following truth table shows the truth values of $\neg(p \vee q)$ (Column 4) and $(\neg p) \wedge (\neg q)$ (Column 7). Since Column 4 and Column 7 are identical in all rows, the equivalence holds.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

As we can see,

$$\neg(p \vee q) \iff (\neg p) \wedge (\neg q).$$

Thus, Law 1 is proven.

Law 2: $\neg(p \wedge q) \iff (\neg p) \vee (\neg q)$.

The following truth table shows the truth values of $\neg(p \wedge q)$ (Column 4) and $(\neg p) \vee (\neg q)$ (Column 7). Since Column 4 and Column 7 are identical in all rows, the equivalence holds.

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

As we can see,

$$\neg(p \wedge q) \iff (\neg p) \vee (\neg q).$$

Thus, Law 2 is proven. □

Quantifiers

We will also need the following symbols, called *quantifiers*:

- The universal quantifier “for every” $\forall$, e.g. $\forall x \in \mathbb{R}, x^2 \geq 0$.
- The existential quantifier “there exists” $\exists$, e.g. $\exists x \in \mathbb{R}, x^2 = 2$.
- “There exists exactly one” $\exists!$, e.g. $\exists! x \in \mathbb{R}, (x \geq 0 \wedge x^2 = 2)$.

Notation: The notation “ $:=$ ” means equality by definition and is used as follows:

defined object $:=$ defining object.

For example, $f(x) := x^2$; the same definition may be written with the defining object on the left, as $x^2 =: f(x)$. Similarly, the notation “ $\Longleftrightarrow$ ” means equivalence by definition, for example

$$A \subseteq B : \Longleftrightarrow \forall x \in A, x \in B.$$

(The expression on the right will be discussed in the next section.)

Throughout, sets are written using the set-builder notation $\{x \in A \mid P(x)\}$, and quantifiers are separated from the statement they govern by a comma, as above.

1.2 Sets

Notice that in order to give an example of a proposition or of a statement involving a quantifier, we needed to refer to some sets or to elements of sets (for instance the integers or the real numbers that you know from school).

The concepts of a set and of membership in a set are *primitive*, i.e. they are meant to be understood intuitively and are not defined. The empty set, i.e. the set to which no element belongs, is denoted by $\emptyset$. The existence of such a set is one of the *axioms* of mathematics, i.e. of the statements we assume to be true without proving them.

Definition 1.2.1 (Set inclusion). *Let A and B be sets. We say that A is a subset of B and write*

$$A \subseteq B : \Longleftrightarrow \forall x \in A, x \in B.$$

We will also write $A \supseteq B$ if $B \subseteq A$.

Definition 1.2.2 (Set equality). *We say that two sets A and B are equal if*

$$A = B : \Longleftrightarrow A \subseteq B \text{ and } B \subseteq A.$$

We write $A \subset B$ if $A \subseteq B$ and $A \neq B$, where $A \neq B : \Longleftrightarrow \neg(A = B)$.

Remark 1.2.3 (Equality by elementwise equivalence). *Let A and B be sets. If for every x we have $x \in A \Longleftrightarrow x \in B$, then $A = B$. Indeed, the implication $x \in A \Rightarrow x \in B$ gives $A \subseteq B$, and the implication $x \in B \Rightarrow x \in A$ gives $B \subseteq A$; the two inclusions together are exactly the definition of equality. We will use this criterion repeatedly below.*

Let A be a set. If a mathematical statement $P(x)$ containing a variable x becomes a proposition whenever we substitute for x an element $a \in A$, we call $P(x)$ a *condition* on A . Such a condition determines the subset of A consisting of all $x \in A$ satisfying P , denoted by

$$\{x \in A \mid P(x)\}.$$

More precisely

$$\{x \in A \mid P(x)\} := \{x \in A \mid P(x) \text{ is true}\}.$$

Example 1.2.4. The statement “2 divides x ” is a condition on $\mathbb{Z}$, and the subset it determines,

$$\{x \in \mathbb{Z} \mid 2 \text{ divides } x\},$$

is the set of even numbers.

Example 1.2.5. The statement $P(x) := (x \leq y^2)$ is not a condition on $\mathbb{R}$, because it contains the extra free variable y . Similarly, the statement $\forall n \in \mathbb{N}, n + 0 = n$ is not a condition on $\mathbb{N}$, because it has no free variable at all.

We can formalise quantifiers in the language of set theory in the following way.

Definition 1.2.6. Let A be a set and let $P(\cdot)$ be a condition on A .

We use the expression

$$\forall x \in A, P(x)$$

to denote the statement

$$\{x \in A \mid P(x)\} = A.$$

We use the expression

$$\exists x \in A, P(x)$$

to denote the statement

$$\{x \in A \mid P(x)\} \neq \emptyset.$$

We use the expression

$$\exists! x \in A, P(x)$$

to denote the statement

$$\{x \in A \mid P(x)\} \text{ has exactly one element.}$$

Exercise 1.2.7. We use the expression

$$\forall! x \in A, P(x)$$

to denote the statement “the condition $P(x)$ holds for all but finitely many x in A ”. Can you formalise it in the language of set theory?

Solution. We use the expression $\forall! x \in A, P(x)$ to denote the statement

$$\{x \in A \mid \neg P(x)\} \text{ is finite.}$$

Example 1.2.8. Let P be an arbitrary condition on the empty set $\emptyset$. We want to determine the truth values of $\forall x \in \emptyset, P(x)$ and of $\exists x \in \emptyset, P(x)$. Since $\{x \in \emptyset \mid P(x)\}$ is a subset of $\emptyset$, we must have

$$\{x \in \emptyset \mid P(x)\} = \emptyset.$$

We conclude that $\forall x \in \emptyset, P(x)$ is true, while $\exists x \in \emptyset, P(x)$ is false.

Definition 1.2.9 (Power set). Let X be a set. The set of all subsets of X is denoted by

$$\mathcal{P}(X) := \{A \mid A \subseteq X\}$$

and is called the power set of X .

Example 1.2.10. For example, $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. If X has N elements, then $\mathcal{P}(X)$ has 2^N elements.

Remark 1.2.11. That $\mathcal{P}(X)$ is again a set is yet another axiom which we need, because of the phenomenon illustrated by Russell's paradox below.

Russell's paradox

Consider the “set” $\mathfrak{R}$ of all sets A satisfying the condition $A \notin A$. If $\mathfrak{R} \in \mathfrak{R}$, then by definition $\mathfrak{R} \notin \mathfrak{R}$. On the other hand, if $\mathfrak{R} \notin \mathfrak{R}$, then $\mathfrak{R} \in \mathfrak{R}$. Because of this paradox, it was concluded that some collections are too big to be sets. Such big collections are called *classes*.

Example 1.2.12. The collection of all sets is a class, and not a set. We denote it by $\mathfrak{U}$.

Definition 1.2.13 (Set difference and complement). If $A, B \subseteq X$, we define the set difference of A and B , and the complement of A in X , by

$$A \setminus B := \{x \in A \mid x \notin B\} \quad \text{and} \quad A^c := X \setminus A$$

respectively. Note that the complement depends on the ambient set X , which is suppressed from the notation; whenever we write A^c , the ambient set must be clear from the context.

Definition 1.2.14 (Set union). Let I and X be sets and let $(A_i)_{i \in I}$ be a family of subsets of X . We call

$$\bigcup_{i \in I} A_i := \{x \in X \mid \exists i \in I, x \in A_i\}$$

the set union of $(A_i)_{i \in I}$. For I of the form $I = \{k, k+1, \dots, N\}$ we write $\bigcup_{i=k}^N A_i$, and for I of the form $I = \{k, k+1, k+2, \dots\}$ we write $\bigcup_{i=k}^{\infty} A_i$.

Exercise 1.2.15. Let X be a set. For I of the form $I = \{1, 2\}$, the set union of sets $A_1, A_2 \subseteq X$ is denoted by $A_1 \cup A_2$. Show that

$$A_1 \cup A_2 = \{x \in X \mid x \in A_1 \vee x \in A_2\}.$$

Solution. Let X be a set and let $A_1, A_2 \subseteq X$. By the definition of set union, for an index set I we have

$$\bigcup_{i \in I} A_i := \{x \in X \mid \exists i \in I, x \in A_i\}.$$

Now take the specific index set $I = \{1, 2\}$. The union of the family $(A_i)_{i \in I}$ is denoted by $A_1 \cup A_2$. Substituting $I = \{1, 2\}$ into the definition gives

$$A_1 \cup A_2 = \bigcup_{i \in \{1, 2\}} A_i = \{x \in X \mid \exists i \in \{1, 2\}, x \in A_i\}.$$

The condition “ $\exists i \in \{1, 2\}, x \in A_i$ ” means that either $i = 1$ works (i.e. $x \in A_1$) or $i = 2$ works (i.e. $x \in A_2$). Hence it is logically equivalent to

$$x \in A_1 \quad \vee \quad x \in A_2.$$

Therefore we obtain

$$A_1 \cup A_2 = \{x \in X \mid x \in A_1 \vee x \in A_2\},$$

which is exactly the desired statement.

Definition 1.2.16 (Set intersection). *Let I and X be sets and let $(A_i)_{i \in I}$ be a family of subsets of X . We call*

$$\bigcap_{i \in I} A_i := \{x \in X \mid \forall i \in I, x \in A_i\}$$

the set intersection of $(A_i)_{i \in I}$. As for unions, we write $\bigcap_{i=k}^N A_i$ when $I = \{k, k+1, \dots, N\}$, and $\bigcap_{i=k}^\infty A_i$ when $I = \{k, k+1, k+2, \dots\}$.

Exercise 1.2.17. *Let X be a set. For I of the form $I = \{1, 2\}$, the set intersection of sets $A_1, A_2 \subseteq X$ is denoted by $A_1 \cap A_2$. Show that*

$$A_1 \cap A_2 = \{x \in X \mid x \in A_1 \wedge x \in A_2\}.$$

Solution. Let X be a set and let $A_1, A_2 \subseteq X$. By the definition of set intersection, for an index set I we have

$$\bigcap_{i \in I} A_i := \{x \in X \mid \forall i \in I, x \in A_i\}.$$

Now choose the specific index set $I = \{1, 2\}$. The intersection of the family $(A_i)_{i \in I}$ is denoted by $A_1 \cap A_2$. Substituting $I = \{1, 2\}$ into the definition yields

$$A_1 \cap A_2 = \bigcap_{i \in \{1, 2\}} A_i = \{x \in X \mid \forall i \in \{1, 2\}, x \in A_i\}.$$

The condition “ $\forall i \in \{1, 2\}, x \in A_i$ ” means that the statement holds for both possible values of i . Thus it is equivalent to requiring that both $x \in A_1$ (when $i = 1$) and $x \in A_2$ (when $i = 2$). In logical notation, this is

$$x \in A_1 \quad \wedge \quad x \in A_2.$$

Therefore we conclude that

$$A_1 \cap A_2 = \{x \in X \mid x \in A_1 \wedge x \in A_2\},$$

which is precisely the statement we set out to prove.

Remark 1.2.18. For the set intersection, if we did not assume that the A_i are all subsets of some fixed set X , we would have to assume that the index set I is non-empty. This is related to Russell's paradox and to conditions on the empty set. Indeed, without an ambient set we would have

$$\bigcap_{i \in \emptyset} A_i = \{x \in \mathfrak{U} \mid \forall i \in \emptyset, x \in A_i\},$$

and by the example above concerning conditions on the empty set, the statement $\forall i \in \emptyset, x \in A_i$ is true for every x . Hence

$$\bigcap_{i \in \emptyset} A_i = \mathfrak{U},$$

which is a proper class and not a set. With the ambient set X fixed, the same computation gives instead $\bigcap_{i \in \emptyset} A_i = X$.

Theorem 1.2.19 (De Morgan's laws for two sets). Let X be a set, and let $A, B \subseteq X$. Then

$$\begin{aligned} (A \cup B)^c &= A^c \cap B^c, \\ (A \cap B)^c &= A^c \cup B^c. \end{aligned}$$

Proof. We prove both laws by showing that for an arbitrary element $x \in X$, the membership conditions on both sides are logically equivalent; by Remark 1.2.3 this suffices to conclude that the sets are equal.

Law 1: $(A \cup B)^c = A^c \cap B^c$.

For any $x \in X$, we have the following chain of logical equivalences:

$$\begin{aligned} x \in (A \cup B)^c &\iff \neg(x \in A \cup B) && \text{(definition of complement)} \\ &\iff \neg(x \in A \vee x \in B) && \text{(definition of set union)} \\ &\iff \neg(x \in A) \wedge \neg(x \in B) && \text{(logical De Morgan)} \\ &\iff (x \in A^c) \wedge (x \in B^c) && \text{(definition of complement)} \\ &\iff x \in A^c \cap B^c && \text{(definition of set intersection).} \end{aligned}$$

Since the equivalence holds for every $x \in X$, the two sets have precisely the same elements. Therefore

$$(A \cup B)^c = A^c \cap B^c.$$

Law 2: $(A \cap B)^c = A^c \cup B^c$.

Similarly, for any $x \in X$:

$$\begin{aligned} x \in (A \cap B)^c &\iff \neg(x \in A \cap B) && \text{(definition of complement)} \\ &\iff \neg(x \in A \wedge x \in B) && \text{(definition of set intersection)} \\ &\iff \neg(x \in A) \vee \neg(x \in B) && \text{(logical De Morgan)} \\ &\iff (x \in A^c) \vee (x \in B^c) && \text{(definition of complement)} \\ &\iff x \in A^c \cup B^c && \text{(definition of set union).} \end{aligned}$$

Again, since this holds for every $x \in X$, we conclude that

$$(A \cap B)^c = A^c \cup B^c.$$

Thus both set-theoretic De Morgan laws follow from the logical De Morgan laws applied to the conditions $x \in A$ and $x \in B$. $\square$

Theorem 1.2.20 (De Morgan's laws for quantifiers). *Let A be a set and let $P(x)$ be a condition on A . Then*

$$\neg(\forall x \in A, P(x)) \iff \exists x \in A, \neg P(x),$$

$$\neg(\exists x \in A, P(x)) \iff \forall x \in A, \neg P(x).$$

Proof. Write

$$S := \{x \in A \mid P(x)\}, \quad S' := \{x \in A \mid \neg P(x)\},$$

so that $S, S' \subseteq A$ and, directly from the definitions of the two conditions,

$$S' = A \setminus S.$$

By definition of the quantifiers, the four statements involved read

$$\forall x \in A, P(x) \text{ is } S = A, \quad \exists x \in A, P(x) \text{ is } S \neq \emptyset,$$

$$\forall x \in A, \neg P(x) \text{ is } S' = A, \quad \exists x \in A, \neg P(x) \text{ is } S' \neq \emptyset.$$

We first record an elementary fact: for a subset $T \subseteq A$,

$$T = A \iff A \setminus T = \emptyset. \quad (*)$$

Indeed, if $T = A$ then $A \setminus T = \{x \in A \mid x \notin A\} = \emptyset$; conversely, if $A \setminus T = \emptyset$ then no element of A lies outside T , i.e. $A \subseteq T$, and since $T \subseteq A$ by assumption, we get $T = A$.

Law 1: $\neg(\forall x \in A, P(x)) \iff \exists x \in A, \neg P(x)$.

By definition, $\neg(\forall x \in A, P(x))$ is the statement $S \neq A$. By $(*)$ applied to $T = S$, this holds if and only if $A \setminus S \neq \emptyset$, that is, if and only if $S' \neq \emptyset$, which by definition is the statement $\exists x \in A, \neg P(x)$.

Law 2: $\neg(\exists x \in A, P(x)) \iff \forall x \in A, \neg P(x)$.

By definition, $\neg(\exists x \in A, P(x))$ is the statement $S = \emptyset$. Now $S = A \setminus S'$, so by $(*)$ applied to $T = S'$ we have $S' = A$ if and only if $A \setminus S' = \emptyset$, i.e. if and only if $S = \emptyset$. Since $S' = A$ is by definition the statement $\forall x \in A, \neg P(x)$, the two are equivalent.

Therefore both De Morgan's laws for quantifiers are proven. $\square$

Theorem 1.2.21 (De Morgan's laws for arbitrary families of sets). *Let I and X be sets and let $(A_i)_{i \in I}$ be a family of subsets of X . Then*

$$\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c, \quad \left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$$

Proof. We prove both equalities by showing that for an arbitrary element $x \in X$, membership in the left-hand side is logically equivalent to membership in the right-hand side; by Remark 1.2.3 this suffices. Throughout the proof we apply De Morgan's laws for quantifiers:

$$\neg(\exists i \in I, P(i)) \iff \forall i \in I, \neg P(i),$$

$$\neg(\forall i \in I, P(i)) \iff \exists i \in I, \neg P(i).$$

Law 1: $\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c.$

For an arbitrary $x \in X$, we have the following chain of equivalences:

$$\begin{aligned}
x \in \left(\bigcup_{i \in I} A_i \right)^c &\iff \neg \left(x \in \bigcup_{i \in I} A_i \right) && \text{(definition of complement)} \\
&\iff \neg (\exists i \in I, x \in A_i) && \text{(definition of set union)} \\
&\iff \forall i \in I, \neg(x \in A_i) && \text{(De Morgan for quantifiers)} \\
&\iff \forall i \in I, x \in A_i^c && \text{(definition of complement)} \\
&\iff x \in \bigcap_{i \in I} A_i^c && \text{(definition of set intersection).}
\end{aligned}$$

Since the equivalence holds for every $x \in X$, the two sets have exactly the same elements. Therefore

$$\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c.$$

Law 2: $\left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$

Again, for an arbitrary $x \in X$, we proceed as follows:

$$\begin{aligned}
x \in \left(\bigcap_{i \in I} A_i \right)^c &\iff \neg \left(x \in \bigcap_{i \in I} A_i \right) && \text{(definition of complement)} \\
&\iff \neg (\forall i \in I, x \in A_i) && \text{(definition of set intersection)} \\
&\iff \exists i \in I, \neg(x \in A_i) && \text{(De Morgan for quantifiers)} \\
&\iff \exists i \in I, x \in A_i^c && \text{(definition of complement)} \\
&\iff x \in \bigcup_{i \in I} A_i^c && \text{(definition of set union).}
\end{aligned}$$

Since this holds for every $x \in X$, we conclude that

$$\left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$$

Thus both De Morgan's laws for arbitrary index sets are proven. $\square$

Definition 1.2.22 (Ordered pair and Cartesian product). *For two sets A and B , we define the Cartesian product of A and B as*

$$A \times B := \{(x, y) \mid x \in A, y \in B\},$$

where the ordered pair is defined by $(x, y) := \{\{x\}, \{x, y\}\}$. For a finite number of sets we set

$$A_1 \times \cdots \times A_n := \{(x_1, \dots, x_n) \mid x_1 \in A_1, \dots, x_n \in A_n\},$$

where the ordered n -tuple is defined recursively for $n \geq 3$ by

$$(x_1, \dots, x_n) := ((x_1, \dots, x_{n-1}), x_n),$$

starting from the ordered pair for $n = 2$. We also write $A^n := \underbrace{A \times \cdots \times A}_{n \text{ times}}.$

Remark 1.2.23. *The recursive definition above is needed because the naive attempt*

$$(x_1, \dots, x_n) := \{\{x_1\}, \{x_1, x_2\}, \dots, \{x_1, \dots, x_n\}\}$$

does not distinguish all tuples: for instance

$$\{\{1\}, \{1\}, \{1, 2\}\} = \{\{1\}, \{1, 2\}\} = \{\{1\}, \{1, 2\}, \{1, 2\}\},$$

so the triples $(1, 1, 2)$ and $(1, 2, 2)$ would be assigned the same set. With the recursive definition one proves, by induction on n using the corresponding property of ordered pairs, that

$$(x_1, \dots, x_n) = (y_1, \dots, y_n) \iff x_i = y_i \text{ for all } i \in \{1, \dots, n\},$$

which is the only property of tuples we shall ever use.

Number sets:

- $\mathbb{N}$ — the set of natural numbers $1, 2, \dots$,
- $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$,
- $\mathbb{N}_k := \{n \in \mathbb{N}_0 \mid n \geq k\}$ for $k \in \mathbb{N}_0$,
- $\mathbb{Z}$ — the set of integers,
- $\mathbb{Q}$ — the set of rational numbers,
- $\mathbb{R}$ — the set of real numbers,
- $\mathbb{C}$ — the set of complex numbers.

We have $\mathbb{N} \subset \mathbb{N}_0 \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. Note that the two definitions of $\mathbb{N}_0$ agree: taking $k = 0$ in the third item gives $\{n \in \mathbb{N}_0 \mid n \geq 0\} = \mathbb{N}_0$, and taking $k = 1$ gives $\mathbb{N}_1 = \mathbb{N}$.

Furthermore, for any of the sets A above we write

$$A^* := A \setminus \{0\},$$

and for A one of $\mathbb{N}_0, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$, which carry an order, we write

$$A_{\geq 0} := \{x \in A \mid x \geq 0\}, \quad A_{> 0} := \{x \in A \mid x > 0\}.$$

The last two notations are not used for $\mathbb{C}$, which is not ordered.

Remark 1.2.24. *The von Neumann natural numbers are defined recursively by:*

$$0 = \emptyset, \quad n + 1 = n \cup \{n\}.$$

Thus,

$$0 = \emptyset, \quad 1 = \{\emptyset\}, \quad 2 = \{\emptyset, \{\emptyset\}\}, \quad 3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}.$$

The set of all natural numbers is the smallest inductive set:

$$\mathbb{N} := \bigcap \{X \mid \emptyset \in X \text{ and } \forall x \in X, x \cup \{x\} \in X\}.$$

1.3 Mathematical Reasoning and Proofs

Truth tables describe the truth values of logically constructed propositions under all possible truth-value assignments. Deduction through mathematical reasoning, by contrast, studies how new propositions can be obtained from propositions already known to be true. In a proof we usually begin with several true propositions as premises and then proceed step by step according to reliable patterns of reasoning.

Theorem 1.3.1. *Let p, q be propositions. If p and $p \Rightarrow q$ are both true, then q is true.*

Proof. Inspect the truth table for $\Rightarrow$. In every row in which both p and $p \Rightarrow q$ are true, q must be true. $\square$

Theorem 1.3.2. *Let p, q, r be propositions. If $p \Rightarrow q$ and $q \Rightarrow r$ are both true, then $p \Rightarrow r$ is true.*

Proof. If p is false, then $p \Rightarrow r$ is true by the definition of a conditional proposition. If p is true, then $p \Rightarrow q$ and Theorem 1.3.1 give that q is true; then $q \Rightarrow r$ gives that r is true. Thus $p \Rightarrow r$ is true. $\square$

Definition 1.3.3 (Direct proof). *To prove a conditional proposition $p \Rightarrow q$, one may assume that p is true and then use this assumption, together with definitions, propositions, and theorems already known, to derive q . This method of proving a conditional proposition is called a direct proof.*

Example 1.3.4. *Let d, n be integers. We say that d divides n and write*

$$d \mid n :\Leftrightarrow \exists k \in \mathbb{Z} \quad n = dk.$$

Prove that if $d \mid n$, then $d \mid n^2$.

Solution. If $d \mid n$, then by the definition of divisibility we can write $n = dk$ for some integer k . Hence

$$n^2 = d^2 k^2 = d(dk^2),$$

so $d \mid n^2$. $\square$

Definition 1.3.5 (Counterexample). *In practice, De Morgan's law for quantifiers*

$$\neg(\forall x \in A, P(x)) \iff \exists x \in A, \neg P(x),$$

means, that to disprove the proposition

$$\forall x \in A, P(x)$$

i.e. to show that its truth value is equal to zero, it is enough to find one element in A that does not satisfy $P(x)$.

Example 1.3.6. *Let n be an integer. The proposition*

$$p : \text{"if } 7 \mid n, \text{ then } 21 \mid n\text{"}$$

should be formally understood as

$$\forall n \in \mathbb{Z} \quad 7 \mid n \Rightarrow 21 \mid n.$$

This proposition is not true, as 70 is divisible by 7 but not divisible by 21.

Definition 1.3.7. Let p, q be propositions. For the conditional proposition $p \Rightarrow q$ we call $p \Rightarrow q$ itself the original proposition. In addition:

- $q \Rightarrow p$ is its converse;
- $(\neg p) \Rightarrow (\neg q)$ is its inverse;
- $(\neg q) \Rightarrow (\neg p)$ is its contrapositive.

Example 1.3.8. Let n be an integer. For the proposition

“if $21 \mid n$, then $7 \mid n$ ”

we write its converse

“if $7 \mid n$, then $21 \mid n$ ”

inverse

“if $21 \nmid n$, then $7 \nmid n$ ”

and contrapositive

“if $7 \nmid n$, then $21 \nmid n$ ”.

We can use concrete integers to determine which of the implications may fail.

Theorem 1.3.9 (Truth-value relation for the contrapositive). Let p, q be propositions. For every choice of truth values of p, q , the original proposition

$$p \Rightarrow q$$

and its contrapositive

$$(\neg q) \Rightarrow (\neg p)$$

have the same truth value. Likewise, the converse $q \Rightarrow p$ and the inverse $(\neg p) \Rightarrow (\neg q)$ have the same truth value.

Proof. Consider the four possible truth-value assignments to p, q :

p	q	$p \Rightarrow q$	$(\neg q) \Rightarrow (\neg p)$	$q \Rightarrow p$	$(\neg p) \Rightarrow (\neg q)$
T	T	T	T	T	T
T	F	F	F	T	T
F	T	T	T	F	F
F	F	T	T	T	T

The third and fourth columns agree, so $p \Rightarrow q$ and $(\neg q) \Rightarrow (\neg p)$ have the same truth value. The fifth and sixth columns agree, so $q \Rightarrow p$ and $(\neg p) \Rightarrow (\neg q)$ also have the same truth value. $\square$

Definition 1.3.10 (Proof by contraposition). Suppose the proposition to be proved is $p \Rightarrow q$. By Theorem 1.3.9, $p \Rightarrow q$ and its contrapositive

$$(\neg q) \Rightarrow (\neg p)$$

have the same truth value. We may therefore replace the task of proving $p \Rightarrow q$ by that of proving $(\neg q) \Rightarrow (\neg p)$: assume $\neg q$ and prove $\neg p$. This method is called proof by contraposition.

Example 1.3.11. Let n be an integer. Prove that if n^2 is even, then n is even.

Solution. We prove the contrapositive: if n is not even, then n^2 is not even. For an integer, not being even means being odd, so

$$n = 2k + 1$$

for some integer k . Therefore

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is again odd. The contrapositive is true, hence so is the original proposition. $\square$

Exercise 1.3.12. Let a, b be real numbers with $a \neq 0$. For arbitrary real numbers x, y , prove by contraposition that if $x \neq y$, then

$$ax + b \neq ay + b.$$

Exercise 1.3.13. Use the truth-value relation between a conditional proposition and its contrapositive to complete the following problems. (1) Suppose $p \Rightarrow q$ and $\neg q$ are both true. Prove that $\neg p$ is true.

(2) Let x be a real number. Prove by contraposition: if $x^2 < 25$, then $-5 < x < 5$.

(3) Let x, y be real numbers. Prove by contraposition: if $xy \neq 0$, then $x \neq 0$ and $y \neq 0$.

(4) Let n be an integer. Prove by contraposition: if $6 \mid n$, then $2 \mid n$ and $3 \mid n$. Try to write the contrapositive in the form “if ... or ..., then ...”.

Exercise 1.3.14. Let x, y be real numbers and consider the propositions

$$p : x^2 + y^2 = 0, \quad q : xy = 0.$$

(1) Prove $p \Rightarrow q$.

(2) Write the converse, inverse, and contrapositive of $p \Rightarrow q$.

(3) Is the converse true? If not, give specific values of x, y .

(4) Which two propositions must have the same truth value? What theorem justifies your answer?

Theorem 1.3.15 (Truth-value relation for an equivalence). Let p, q be propositions. For every choice of truth values of p, q , the proposition

$$p \Leftrightarrow q$$

and the proposition

$$(p \Rightarrow q) \wedge (q \Rightarrow p)$$

have the same truth value.

In particular, to prove $p \Leftrightarrow q$, it suffices to prove separately

$$p \Rightarrow q \quad \text{and} \quad q \Rightarrow p.$$

Proof. Consider the four truth-value assignments to p, q . By the truth rules for equivalence and implication, the third and final columns in the table below agree in every row. Hence $p \Leftrightarrow q$ and $(p \Rightarrow q) \wedge (q \Rightarrow p)$ always have the same truth value.

p	q	$p \Leftrightarrow q$	$p \Rightarrow q$	$q \Rightarrow p$	$(p \Rightarrow q) \wedge (q \Rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

If $p \Rightarrow q$ and $q \Rightarrow p$ have both been proved, then their conjunction $(p \Rightarrow q) \wedge (q \Rightarrow p)$ is true. By the truth-value relation just established, $p \Leftrightarrow q$ is also true. Thus an equivalence may be proved by establishing its two directions separately. $\square$

Corollary 1.3.16. *As a consequence, for sets A, B , instead of proving $A = B$ directly we can prove that $A \subseteq B$ and $B \subseteq A$.*

Exercise 1.3.17. *In each item below, a value of the real number x is specified. Let p, q be the two propositions indicated. First determine their truth values, and then use the definition of the equivalence to determine the truth value of $p \Leftrightarrow q$.*

- (1) Let $x = 2$, $p : x = 2$, $q : x^2 = 4$.
- (2) Let $x = -4$, $p : x = -4$, $q : |x| = 4$.
- (3) Let $x = 4$, $p : x = -4$, $q : |x| = 4$.
- (4) Let $x = 5$, $p : |x| < 3$, $q : x^2 < 9$.

Exercise 1.3.18. *Let x be a real number. Prove*

$$x^2 - 7x + 12 = 0 \Leftrightarrow x = 3 \text{ or } x = 4.$$

Split the equivalence into two directions and prove them separately. Indicate the main algebraic operation used in each direction.

1.3.1 Proof by Contradiction

A direct proof normally starts from known propositions and proceeds step by step toward the desired conclusion. In some problems, however, the negation of the target proposition has a more concrete algebraic or geometric meaning, and combining it with what is already known quickly leads to an impossible conclusion. This motivates proof by contradiction.

The logical core of proof by contradiction is the following: if assuming that a proposition is false forces us, within the same chain of reasoning, to obtain both some proposition r and its negation $\neg r$, then the assumption cannot be correct. The next definition calls such a situation a contradiction and formulates the method precisely. Unlike proof by contraposition, proof by contradiction does not require the target to have been written in advance as a particular conditional proposition, so it applies naturally to irrationality, nonexistence, uniqueness, and many other problems.

Definition 1.3.19. *If a collection of assumptions allows us to derive both a proposition r and its negation $\neg r$, we say that the assumptions lead to a contradiction.*

To prove a proposition p , assume first that $\neg p$ holds. If this assumption, together with propositions already known to be true, leads to a contradiction, then $\neg p$ cannot hold, and therefore p holds. This method is called proof by contradiction.

Exercise 1.3.20. *When proving $p \Rightarrow q$, compare proof by contraposition and proof by contradiction.*

- (1) *What do we assume in a proof by contraposition, and what is the goal?*
- (2) *What do we assume in a proof by contradiction? Why can that assumption be written as the simultaneous assumptions p and $\neg q$?*

Example 1.3.21. *Prove that $\sqrt{2}$ is irrational.*

Proof. (By so-called infinite descent method) Assume that $\sqrt{2}$ is rational. Write it in lowest terms as

$$\sqrt{2} = \frac{a}{b},$$

where a, b are positive integers with no common divisor greater than 1. Squaring gives

$$a^2 = 2b^2.$$

Thus a^2 is even. By the result proved earlier, a is even, so $a = 2c$ for some integer c . Substituting this back gives

$$4c^2 = 2b^2,$$

and hence $b^2 = 2c^2$, so b is even as well. Thus both a and b are divisible by 2, contradicting the assumption that a/b is in lowest terms. Therefore the original assumption is false, and $\sqrt{2}$ is irrational. $\square$

Exercise 1.3.22. *The following proof by contradiction is incomplete. Fill in the missing logical steps.*

Let x be a real number. Prove: if $x^2 < 9$, then $x < 4$.

Proof: Assume $x^2 < 9$ but $x \geq 4$. From $x \geq 4$ we obtain _____. This contradicts _____. Hence $x < 4$.

Exercise 1.3.23. *Let a, b, c be positive real numbers satisfying*

$$a + b + c = 2.$$

Use proof by contradiction to show that if $a > \frac{3}{4}$ and $b > \frac{3}{4}$, then $c < \frac{1}{2}$.